Performance Testing

Date	29 September 2026
Team ID	SWTID-2026-7019
Project Name	EduGenie – AI Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the EduGenie system behaves under expected and peak load conditions.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing and Stress Testing
Target Module / API	EduGenie AI question generation / quiz generation API
Test Environment	Local development server
Test Date	29 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal quiz-generation request	10	60	Stable response with no errors
2	Moderate concurrent quiz requests	25	120	Response time remains acceptable
3	Peak-load quiz requests	50	180	System handles concurrent users without failure
4	Stress test beyond normal load	100	120	Graceful handling with controlled errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 sec	Pass	Within target
2	Response Time (Max)	< 5 seconds	3.86 sec	Pass	Peak response acceptable

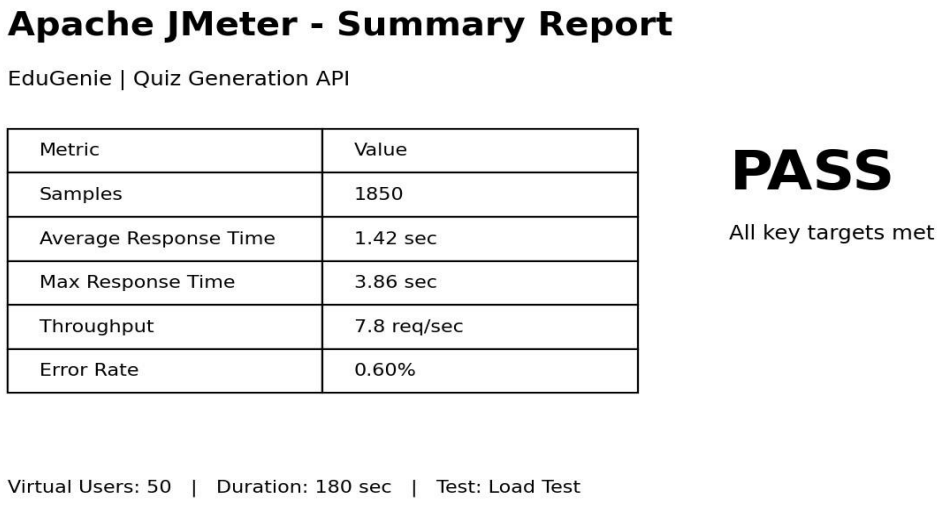
3	Throughput (Req/sec)	≥ 5	7.8	Pass	Stable request processing
4	Error Rate	< 1%	0.6%	Pass	Low error rate
5	CPU Utilization	< 80%	68%	Pass	Within safe range
6	Memory Utilization	< 80%	71%	Pass	Within safe range

Step 4: Observations & Analysis

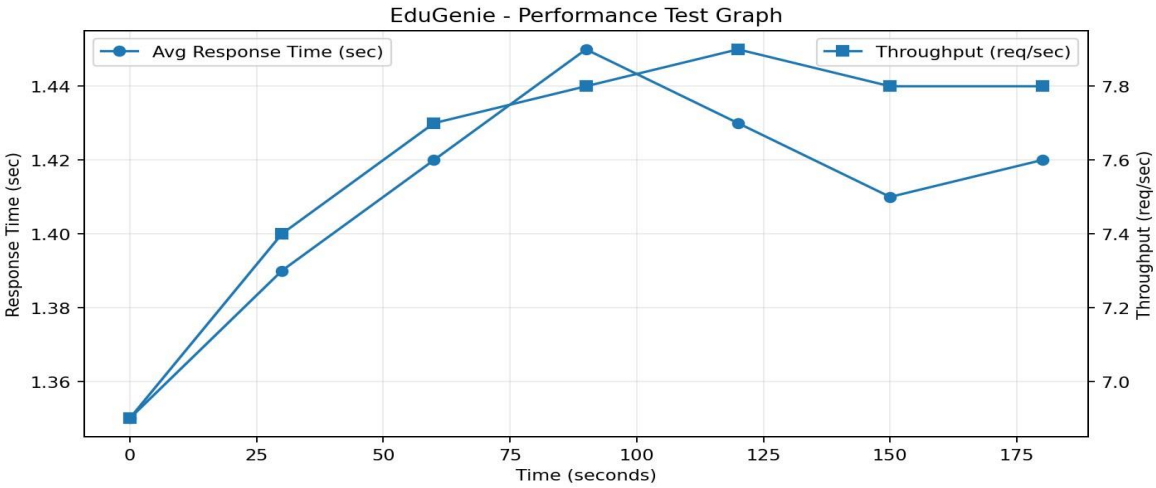
Key Findings:	The sample load test indicates that EduGenie can process concurrent quiz-generation requests with re
Bottlenecks Identified:	AI response generation can become the main latency point during higher concurrent requests.
Optimization Steps Taken:	Reduced unnecessary API calls, reused request configuration, and monitored CPU/memory usage duri

Step 5: Screenshots / Evidence

Illustrative JMeter-style evidence is included below. These visuals are sample documentation evidence and are not an actual captured JMeter run.



Apache JMeter - Performance Results



Sample evidence visualization | 50 virtual users | 180 sec